

Lab Nov 17

Matrices in R

Let's use our WHI experiment data

```
whi<- read.csv("whidata.csv")
whi$age50s<- ifelse(whi$age=="50-59", 1, 0)
whi$age60s<- ifelse(whi$age=="60-69", 1, 0)
whi$age70s<- ifelse(whi$age=="70-79",1,0)
```

Start with simple additive model

```
whimod1<- lm(breastcancer ~ treat + age60s + age70s, data=whi)

summary(whimod1)
```

Call:

```
lm(formula = breastcancer ~ treat + age60s + age70s, data = whi)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.02823	-0.02539	-0.02028	-0.02006	0.98505

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.014945	0.002281	6.552	5.85e-11 ***
treat	0.005112	0.002263	2.259	0.02391 *
age60s	0.005337	0.002585	2.065	0.03895 *
age70s	0.008168	0.003129	2.611	0.00904 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1458 on 16604 degrees of freedom
 Multiple R-squared: 0.0007668, Adjusted R-squared: 0.0005863
 F-statistic: 4.247 on 3 and 16604 DF, p-value: 0.005241

How can we get the same thing using matrices? We want $(X'X)^{-1}X'Y$, so we need a matrix X that has our covariates in it and a column of 1s so we can estimate an intercept.

```
df.mat<- data.frame(constant=rep(1,length(whi$breastcancer)), treat=whi$treat, age60s=whi$age60s)

X<- data.matrix(df.mat)

dim(X)
```

```
[1] 16608      4
```

To test our matrix functions, I want to use something smaller to be able to check it. I will use

the matrix from PS 3: $C = \begin{bmatrix} 3 \\ 2 \\ 5 \\ 7 \end{bmatrix}$ and $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$

```
A<- matrix(c(1,1,1,1,0,1,1,2),ncol=2)
dim(A)
```

```
[1] 4 2
```

```
C<- matrix(c(3,2,5,7),ncol=1)
dim(C)
```

```
[1] 4 1
```

```
#What is the transpose of A?
t(A)
```

```
      [,1] [,2] [,3] [,4]
[1,]     1     1     1     1
[2,]     0     1     1     2
```

```
t(A)%*%A
```

```
      [,1] [,2]  
[1,]     4     4  
[2,]     4     6
```

```
#What is the inverse of A'A?  
solve(t(A)%*%A)
```

```
      [,1] [,2]  
[1,]  0.75 -0.5  
[2,] -0.50  0.5
```

```
solve(t(A)%*%A)%*%t(A)%*%C
```

```
      [,1]  
[1,]  2.25  
[2,]  2.00
```

Let's go back to our WHI data

```
betas<- solve(t(X)%*%X)%*%t(X)%*%whi$breastcancer  
dim(betas)
```

```
[1] 4 1
```

```
betas
```

```
      [,1]  
constant 0.014945101  
treat     0.005112105  
age60s    0.005336953  
age70s    0.008167928
```

We got the same answer as lm!

optional: make a new version of X that includes all three age variables. What happens?

```
df.mat2<- data.frame(constant=rep(1,length(whi$breastcancer)), treat=whi$treat, age50s=whi$age50s)

X2<- data.matrix(df.mat2)

dim(X2)
```

```
[1] 16608      5
```

```
#solve(t(X2)%*%X2)
```

How do we use matrices to estimate standard errors for our betas? What is the sandwich estimator?

$$\hat{V}[\hat{\beta}] = [X^T X]^{-1} X^T \text{diag}[(Y - X\hat{\beta})^2] X [X^T X]^{-1}.$$

We know we can deal with X now, but what is the thing in the middle of the sandwich? A diagonal matrix of the residuals squared.

```
resid<- residuals(whimod1)
```

so we have our residuals and we can square them, but how do we make a diagonal matrix?

```
test<- c(4,5,6,7,8)

diag(test)
```

```
      [,1] [,2] [,3] [,4] [,5]
[1,]    4    0    0    0    0
[2,]    0    5    0    0    0
[3,]    0    0    6    0    0
[4,]    0    0    0    7    0
[5,]    0    0    0    0    8
```

So we can make a diagonal matrix of our squared residuals the same way. What should the dimensions of the matrix be?

```
D<- diag(resid^2)

dim(D)
```

```
[1] 16608 16608
```

Let's use our sandwich estimator

```
betavar<- solve(t(X)%*%X)%*%t(X)%*%D%*%X)%*%solve(t(X)%*%X)
```

Are we done? The diagonal is the variance of our betas and we need to take the squareroot to get the standard error.

```
sqrt(diag(betavar))
```

	constant	treat	age60s	age70s
	0.002061465	0.002257021	0.002471209	0.003181865

That was complicated. Big matrices. Is there an easier way to do it? The sandwich package

```
library(sandwich)  
sandwich(whimod1)
```

	(Intercept)	treat	age60s	age70s
(Intercept)	4.249638e-06	-2.401299e-06	-3.154447e-06	-3.273013e-06
treat	-2.401299e-06	5.094142e-06	5.382982e-08	2.871100e-07
age60s	-3.154447e-06	5.382982e-08	6.106872e-06	3.127476e-06
age70s	-3.273013e-06	2.871100e-07	3.127476e-06	1.012427e-05

```
sqrt(diag(sandwich(whimod1)))
```

	(Intercept)	treat	age60s	age70s
	0.002061465	0.002257021	0.002471209	0.003181865